C Vertical and horizontal datasets

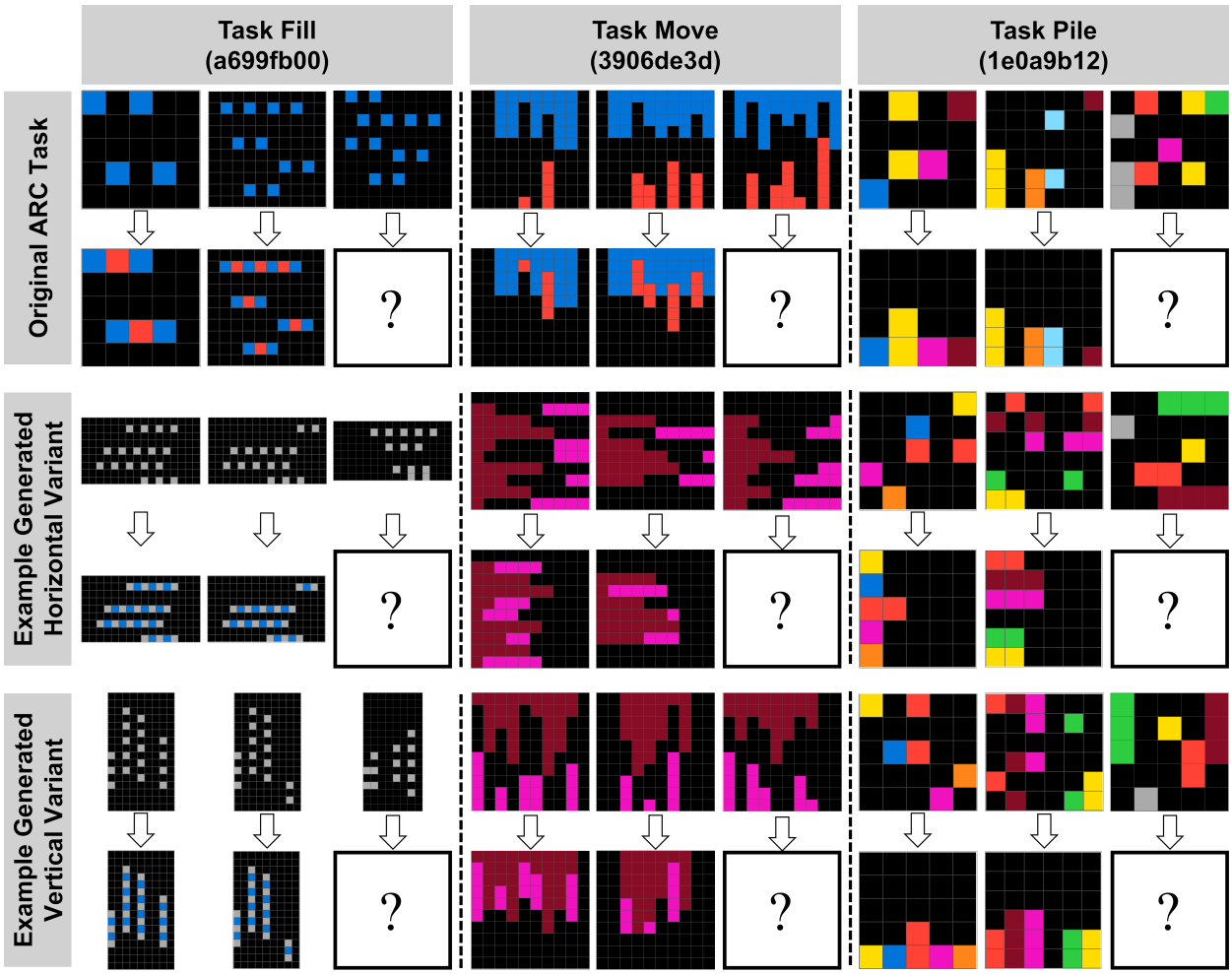

Figure 12: **Visualization for generated vertical and horizontal variants for three ARC tasks.** Each column displays the original ARC task on the top, one generated horizontal variant in the middle, and one generated vertical variant at the bottom.

D GPT reasonings for solved tasks

Table 6: **GPT Reasoning for correctly solved ARC tasks using best performing direct-grid approach.** The correctness of the reasoning provided is determined manually. If a reasoning was not provided, it is reported as N/A.

Task	Reasoning	Correctness
25ff71a9	Transformation applied: 1. Move color objects (blue or red) 1 row down.	✓
aabf363d	N/A	✗

d5d6de2d	Transformation applied: 1. Replace color red object with color green object 2. Remove the outer border of the color green object	✓
dc433765	Transformation applied: 1. Move color green object 1 pixel down and color yellow object 1 pixel up.	✗
b1948b0a	N/A	✗
74dd1130	N/A	✗
c0f76784	Transformation applied: 1. Fill the empty spaces inside the gray objects with the following colors: - Cyan for the top object - Orange for the middle object - Purple for the bottom object	✗
6150a2bd	N/A	✗
c8f0f002	N/A	✗
67385a82	N/A	✗
67a3c6ac	N/A	✗
a5313dff	Transformation applied: 1. Replace the black cells inside the red shape with blue cells.	✓
a699fb00	Transformation applied: 1. Add a red object to the right of every blue object, except when there is already a red object to the right of the blue object.	✗

Table 7: **GPT Reasoning for correctly solved ARC tasks using best performing object-based approach.** The correctness of the reasoning provided is determined manually. If a reasoning was not provided, it is reported as N/A.

Task	Reasoning	Correctness
a79310a0	Transformation applied: 1. Move color 8 object 1 pixel to the right 2. Recolor color 8 object to color 2	✓
25ff71a9	Transformation applied: 1. Move the object 1 pixel to the right.	✓
aabf363d	Transformation applied: 1. Recolor the larger object with the color of the smaller object.	✓
d5d6de2d	Transformation applied: 1. Remove color-2 objects 2. Recolor color-0 objects to color-3	✓
6c434453	Transformation applied: 1. For each color-1 object with size 8, remove the corners to create two color-2 objects with size 5.	✓

dc1df850	Transformation applied: 1. Surround each color-2 object with color-1 objects in a 3x3 square, excluding the center where the color-2 object is located.	✓
aedd82e4	Transformation applied: 1. Recolor all color-2 objects that are not touching the largest color-2 object to color-1.	✓
dc433765	Transformation applied: 1. Move color 3 object 1 pixel towards color 4 object.	✓
6e82a1ae	Transformation applied: 1. Recolor color 5 objects to color 1 if size is 4 2. Recolor color 5 objects to color 2 if size is 3 3. Recolor color 5 objects to color 3 if size is 2	✓
b1948b0a	Transformation applied: 1.Recolor color 6 objects to color 2	✓
4258a5f9	Transformation applied: 1. For each color-5 object, create a color-1 object with 8 pixels surrounding the color-5 object.	✓
ded97339	Transformation applied: 1. Connect color-8 objects with the same color in a straight line.	✓
7f4411dc	Transformation applied: 1.Remove all color objects with size 1.	✓
e9614598	Transformation applied: 1.Find the midpoint between the two color-1 objects. 2.Create a color-3 object of size 5 in a cross shape centered at the midpoint.	✓
ea32f347	Transformation applied: 1. Recolor the largest color-5 object to color-1 2. Recolor the smallest color-5 object to color-2 3. Recolor the remaining color-5 object to color-4	✓
6150a2bd	Transformation applied: 1.Rotate the colors of the objects 90 degrees clockwise.	✗
913fb3ed	Transformation applied: 1. For each object, create a new object with 8 coordinates surrounding the original object. 2. Assign colors to the new objects based on the original object's color: color 2 -> color 1, color 3 -> color 6, color 8 -> color 4.	✓
c8f0f002	Transformation applied: 1.Recolor color 7 objects to color 5	✓
67385a82	Transformation applied: 1.Recolor color 3 objects with size > 1 to color 8.	✓
ce22a75a	Transformation applied: 1. Surround each color-5 object with color-1 objects in a 3x3 square, keeping the color-5 object in the center.	✗